

Sixteen proofs, five kinds of correctness

Audit of **mathlib-proofs-16.zip**, submitted as foundational machinery for formalizing Nassim Nicholas Taleb's Statistical Consequences of Fat Tails.

Verdict

The package contains useful mathematical ingredients, but it is **not ready to be accepted as sixteen verified formalizations**. Proof12 and Proof14 state false equalities because they interchange location and scale. Proof09 also elaborates to a false arbitrary-domain claim because of an unqualified name. Several other true statements have Lean proof defects. Some successful analytic identities are described more broadly than their formal conclusions justify.

Reference build: 8 of 16 original files passed and 8 failed under the pinned audit environment. Passing files: **01, 04, 07, 10, 11, 13, 15, 16**. This is a result about these exact submitted files on the stated version, not a claim about an unavailable original project.

The findings that matter most

Three false statements: Proof12 and Proof14 interchange location and scale. Proof09 accidentally quantifies over every real set instead of the universal set. Their intended corrected statements are true.

Incorrect stable-law model: the tangent-only expression loses skewness at $\alpha = 1$ and allows parameters that cannot describe a probability law. Proof16's algebraic identity must be separated from its probabilistic interpretation.

Proof errors: reversed limit composition, an inappropriate real-power continuity argument, an unreconciled exponent order, reversed eventual equality, and misuse of affine composition. Compiler details are recorded individually below.

Definition and scope gaps: the proposed subexponential predicate does not connect its two tails by convolution; the real-valued tail exponent mishandles zero tails; the Fréchet expression is not a global CDF.

Positive findings: the elementary limit, scaling, and exponential identities are mostly mathematically sound. The report credits those results individually and separates optional shorter proofs from mandatory corrections.

A companion repair bundle contains the checked repair modules, the pinned build configuration, original-file compiler logs, and instructions. It does not claim to complete the remaining probability-measure formalization.

What was checked

Evidence and integrity

The supplied ZIP matches the user-provided SHA-256 exactly. Its internal SHA256SUMS also validates all sixteen Lean files and INDEX.md. The archive contains those eighteen files only: no Lake project, lean-toolchain, dependency lockfile, CI log, or original FatTails/Machinery source tree.

```
b0132c09a8835107abaac916578e5e61a703  
b5a09a800c2e874c24d15d05b47c
```

The hash is printed on two lines for readability; concatenate them without whitespace. No original proof file was edited during testing.

Four separate questions for every file

Statement: Is the proposition actually written true? **Proof:** Do the supplied Lean commands establish it? **Meaning:** Do the definitions and docstrings encode the intended mathematics? **Scope:** Does the theorem establish a distribution-level result or only an analytic ingredient? Elegance is assessed after those questions.

Reproducible reference environment

Lean **4.24.0**; Mathlib tag **v4.24.0**, commit **f897ebcf72cd16f89ab4577d0c826cd14afaafc7**. The audit chose this explicit stable reference because the pack supplies no target version. Exact original files were compiled separately. A failure attributable to this baseline is labeled as such; a mathematical counterexample is independent of version.

Repair modules were checked in the same environment. The broadest import was avoided; additional imports are explicit where the repairs require them. The companion includes the complete diagnostics rather than presenting speculative source review as compiler output.

All three repair modules compile with zero errors and zero warnings. The 29 exported repair and diagnostic theorems were checked with `#print axioms`: each depends only on `propext`, `Classical.choice`, and `Quot.sound`; none reports `sorryAx`. Two exact counterexamples to the original Gaussian/Cauchy specializations are themselves Lean-checked. The corrected piecewise S_1 expression's scaling theorem is also checked. A separate diagnostic checks Proof09's captured implicit set and disproves its resulting statement. The final four-module companion project also passes a genuine aggregate Lake build.

Trust and scope limitations

A successful Lean run checks a theorem relative to its imported definitions and axioms; it does not validate the English interpretation automatically. Conversely, a broken script does not make its theorem false. The submitted source contains no explicit `sorry`, `admit`, or `new axiom`, but absence of those strings alone is not a build certificate.

The audit reviews all sixteen supplied files deeply. It is not a certification of the missing parent project, a Mathlib acceptance decision, or a comprehensive audit of the entire book. Source comparisons use arXiv v4, the 2025 third revised edition. [R1]

Per-proof verdict matrix

“True” refers to the literal analytic proposition under its local definitions. The build column describes the original file on the audit baseline. Semantic limitations remain relevant even when a file passes. Detailed findings and line references follow.

Proof	Mathematics	Original build	Principal action
01	True	PASS	Keep argument; simplify logarithmic algebra
02	True	FAIL	Remove no-op simp; simplify proof
03	True	FAIL	Reverse Tendsto composition order
04	True	PASS	Keep product-of-ratios argument
05	True	FAIL	Fix continuity API and exponent order
06	True	FAIL	Remove unreachable ring; factor proof
07	True finite limit	PASS	Keep limit; repair general exponent API
08	True conditional limit	FAIL	Fix congruence direction; rename relation
09	FALSE as elaborated	FAIL	Qualify Set.univ; fix composition
10	True raw identity	PASS	Keep algebra; fix global CDF/support
11	True	PASS	Keep positive-size Gumbel identity
12	FALSE	FAIL	Swap location and scale arguments
13	True	PASS	Keep Gaussian scaling identity
14	FALSE	FAIL	Swap location and scale arguments
15	True algebra	PASS	Keep; qualify probability interpretation
16	True expression identity	PASS	Keep algebra; fix stable-law semantics

Fully correct examples: Proof01, Proof04, Proof11, Proof13, and Proof15 pass unchanged and are mathematically complete for their stated analytic identities. Proof11 is particularly clean: no mathematical or tactic repair is needed. Proof07, Proof10, and Proof16 also prove their literal analytic statements correctly; their broader definitions or probability interpretations need the qualifications detailed below. Packaging concerns do not erase this positive credit.

Issues shared across the package

The “verified” claim is not supported by the delivered bytes

INDEX.md:3-5 calls the files self-contained and says the proof bodies are verified. The false statements rule out an all-successful, axiom-free check of these exact statements in a consistent environment. An extraction mistake, argument-order change, or stale verification may explain the discrepancy. The evidence does not identify which occurred, and it does not justify accusing the author of dishonesty.

Standalone snippets are not composable modules

IsSlowlyVarying is redeclared globally in Proof01-06, IsRegularlyVarying in Proof04-06, and stableCharFn in Proof12, Proof14, and Proof16. Importing them as one ordinary library produces declaration collisions. Centralize definitions and give the project a namespace; retain one declaration of each predicate or function. Identical text does not make duplicate declarations merge.

Total functions are not implicit domain assumptions

Lean defines division by zero, Real.log at zero, tangent at its classical poles, and real powers outside their ordinary positive-base setting. Those extensions make expressions well-typed everywhere. They do not make every expression a survival function, CDF, or characteristic function. Most issues in Proof07, Proof10, and Proof16 concern this boundary between algebra and interpretation.

Ratio properties versus classical regular variation

The local predicates encode ratios only. They deliberately admit negative constants and do not impose measurability or a Baire condition. The elementary lemmas audited here remain valid. Stronger regular-variation results may need additional assumptions. A useful design is a small ratio-limit predicate plus separately named positivity and regularity hypotheses; do not silently apply classical representation or uniform-convergence theorems to the weaker predicate.

Probability bridges remain explicit work

For maxima, one needs a theorem turning independence into a product of CDFs. For sums, one needs probability measures or random variables, independence, the characteristic-function product theorem, and uniqueness of laws. A symbolic expression called “CDF” or “characteristic function” supplies none of these automatically. The bundle preserves useful analytic work while making these missing conclusions explicit.

Logarithm is slowly varying

Mathematically correct; the eventual-domain argument is sound.

Proof01_SlowlyVaryingLog.lean:23-47

For $b > 0$ and $x > 1$, both the logarithm identity and cancellation are legitimate: $\log(bx)/\log x = 1 + \log b/\log x$. Since $\log x$ tends to infinity, the correction tends to zero. The proof correctly works eventually, so $x = 1$ and nonpositive x do not create a missing assumption.

The use of `e.symm` in the final `Tendsto.congr'` is correct: the proof has convergence of the simplified expression and transfers it back to the original ratio. This direction matters, because Proof08 gets the analogous step wrong.

Lean result

PASS unchanged, with no warnings. The suggested direct rewriting proof also compiles. This file is a fully correct elementary lemma under its stated ratio definition.

Optional simplification

Replace broad field normalization by the exact algebra being used:

```
rw [Real.log_mul (ne_of_gt hb) (ne_of_gt hx0),
    add_div, div_self hlogx, add_comm]
```

This is easier to read and makes the nonzero denominator visible. The repaired module uses this direct proof and retains the same mathematical statement.

Credit and scope

The mathematical proof needs no change. This is a useful basic lemma, not a Tauberian or integrability theorem. It also supplies a counterexample to any claim that slow variation alone forces convergence to a finite constant: $\log x$ is slowly varying but unbounded.

Nonzero constants are slowly varying

Correct statement and proof idea; the nonzero hypothesis is necessary.

Proof02_SlowlyVaryingConst.lean:20-35

For every scaling b , the ratio of a nonzero constant to itself is exactly 1. The proof's eventual equality is valid everywhere, so the convergence follows immediately. The unused positivity assumption on b comes from unpacking the definition, not from a missing mathematical step.

The condition $c \neq 0$ is substantive: for the zero function, Lean's ratio $0/0$ is 0, so the defining limit would be 0 rather than 1. Negative constants qualify under the package's ratio-only definition. This is consistent with the formal statement, although conventional positive-function terminology needs care.

Lean result

FAIL on the reference baseline: line 33, `simp only []`, reports 'simp made no progress'. This is a local tactic defect, not a false theorem. Delete that no-op step, or use the checked shorter proof below.

Shorter checked approach

```
intro b hb
simp only [div_self hc] using
  (tendsto_const_nhds :
    Tendsto (fun _ : Real => (1 : Real)) atTop (nhds 1))
```

The intermediate eventual-equality proof, `filter_upwards`, and `simp only []` can be removed. The statement needs no mathematical correction. In the tested version, removing the no-op `simp` step is mandatory for the original script to compile; shortening the rest is optional.

Credit

Within the stated definition this is a complete elementary result. No missing domain case or counterexample was found. Its correctness should be retained as a positive finding even though the shared packaging needs consolidation.

A nonzero finite limit implies slow variation

True theorem; definite composition-order error in the submitted proof.

Proof03_SlowlyVaryingOfTendsto.lean:28-37, especially line 33

The mathematical chain is $bx \rightarrow +\infty$, followed by $L(bx) \rightarrow c$, followed by division of two limits tending to the same nonzero c . The submitted line `hmul.comp hL` reverses the first two maps. `Tendsto.comp` takes the outer convergence first: the outer function is L and the inner function is multiplication by b .

Mandatory correction

```
-- Submitted:
have h1 : Tendsto (fun x => L (b * x)) atTop (nhds c) :=
  hmul.comp hL

-- Correct:
have h1 : Tendsto (fun x => L (b * x)) atTop (nhds c) :=
  hL.comp hmul
```

The wrong order does not merely describe the wrong composite: the intermediate filters do not match. L tends to a neighborhood of c , whereas `hmul` expects an input approaching `atTop`. [R2]

Lean result

FAIL at line 33 with a `Tendsto.comp` type mismatch. The corrected composition and shortened proof compile without warnings.

Compact ending

```
simp only [div_self hc] using (hL.comp hmul).div hL hc
```

This theorem is a sufficient condition for slow variation, not a characterization. Proof01 shows why the converse fails. Preserve the quantifier “nonzero” in both the theorem and its documentation.

Products add regular-variation indices

Correct elementary closure law; no positivity assumption is missing.

Proof04_RegularlyVaryingMul.lean:27-45

The key identity is $(A \cdot B) / (C \cdot D) = (A/C) \cdot (B/D)$. It holds in Lean's field semantics even if C or D is zero. Therefore this proof does not illegitimately cancel a potentially zero function value. Multiplying the two ratio limits and using $b^{(\alpha_1 + \alpha_2)} = b^{\alpha_1} \cdot b^{\alpha_2}$ for $b > 0$ yields the stated index.

Lean result

PASS unchanged, with no warnings. The shorter replacement also compiles. No mathematical repair is needed.

Optional simplification

```
intro b hb
simp only [div_mul_div_comm, Real.rpow_add hb] using
  (h1 b hb).mul (h2 b hb)
```

Here h1 and h2 stand for the two regular-variation hypotheses. The companion module contains the exact compiled version. The submitted eventual-equality block is harmless but unnecessary because the algebraic equality holds everywhere.

Credit and scope

The statement and reasoning are sound under the package's definitions. There is no need to invent additional positivity requirements to rescue this result. However, it concerns multiplication of functions, not multiplication of random variables or the distribution of a product. Those are different propositions and would need separate probability arguments.

Powers multiply regular-variation indices

True theorem; one missing import and two further proof defects.

Proof05_RegularlyVaryingRpow.lean:35-52, especially lines 50-52

Eventual positivity makes the quotient-of-powers identity valid. The limiting base b^α is strictly positive, so raising the ratio to any fixed real p is continuous there, including $p = 0$ and $p < 0$. The desired limit is $(b^\alpha)^p = b^{(\alpha p)} = b^{(p\alpha)}$.

Mandatory repairs

Import `Pow.Continuity` first. Then `Tendsto.rpow` does not accept eventual nonnegativity as its last premise on the checked API. It requires a condition on the limiting base and exponent. The fixed-exponent `rpow_const` lemma expresses the intended reasoning directly. Also, $\alpha * p$ and $p * \alpha$ are not definitionally equal: the final exact command cannot silently commute them. [R3]

```
have hlim := (hL b hb).rpow_const
  (Or.inl (ne_of_gt (Real.rpow_pos_of_pos hb alpha)))
-- Rewrite the limiting value, explicitly using commutativity:
-- (b ^ alpha) ^ p = b ^ (alpha * p) = b ^ (p * alpha)
```

The displayed identifiers `alpha` and `hL` are schematic; the companion provides the complete checked theorem with its actual binders. It imports `Pow.Continuity` explicitly. The redundant `hnn` proof can be deleted, while the eventual positivity facts used for `Real.div_rpow` should remain.

Lean result

FAIL: the original import list does not expose the required `rpow` limit lemma. A controlled run adding `Pow.Continuity` then reports the wrong `hnn` argument type. After fixing that premise, another controlled run reaches the $\alpha * p$ versus $p * \alpha$ mismatch. The complete replacement fixes all three issues and compiles without warnings.

Do not overcorrect

Dropping positivity indiscriminately is unsafe: the total real-power extension on negative bases is not the ordinary positive-base power law. This differs from Proof06, whose positivity assumption can genuinely be removed.

Normalization by a power gives slow variation

Correct equivalence; a substantially cleaner general theorem is available.

Proof06_RViffSlowlyVarying.lean:36-74

For $x > 0$ and $b > 0$, the normalized ratio equals the original ratio divided by b^α . The forward limit is therefore $b^\alpha/b^\alpha = 1$, and the reverse direction multiplies a limit of 1 by b^α . Both directions are mathematically valid.

$$\begin{aligned} & (L (b*x) / (b*x)^\alpha) / (L x / x^\alpha) \\ &= (L (b*x) / L x) / b^\alpha \end{aligned}$$

Lean result

FAIL at lines 54 and 71: field_simp has already closed the goals, so each following ring reports 'No goals to be solved'. Deleting those redundant steps is the minimal baseline repair. The companion also supplies the stronger, factored theorem described below.

Useful simplifications

The shifted positivity assumption hpos' is introduced twice and never needed. $x > 1$ can be weakened to $x > 0$. More strongly, the whole hpos assumption can be removed: if $L x = 0$ then both sides of the displayed identity are zero; otherwise the usual field calculation applies. Only x^α and b^α need to be nonzero.

The checked repair proves one normalization identity and reuses it in both directions. It removes the positivity hypothesis instead of duplicating field algebra. This is a generalization of a valid theorem, not a repair of a false statement.

Scope and attribution

The index α is already assumed in the regular-variation predicate. This proof does not derive a power-law ratio limit from an unknown multiplicative limit function, and it is not the full Karamata representation theorem. The Bingham-Goldie-Teugels theorem-number citation should be checked against the intended edition before publication; this audit does not certify that bibliographic pinpoint.

Exact Pareto log-tail exponent

The finite-limit statement is correct; the surrounding exponent definition needs care.

Proof07_TailExponentPareto.lean:21-44

For $x > 1$, $-\log(Cx^{(-\alpha)})/\log x = \alpha - \log C/\log x$. With $C > 0$, the correction tends to zero. The analytic identity holds for every real α ; $\alpha > 0$ is needed for the intended decaying Pareto-tail interpretation, not for this limit calculation.

Lean result

PASS unchanged. The only warning is the unused $h\alpha$ hypothesis, consistent with the stronger analytic result. The repaired finite-exponent predicate, Pareto equality corollary, and legacy-definition diagnostics all compile without warnings.

The original anonymous lambda placeholders elaborate successfully in this build; they are not a syntax error. Naming x is merely clearer. The repair also states the missing corollary: the finite real-valued $\liminf$ equals α , obtained from the proved convergence by $\liminf_eq$.

Definition-level problem

`tailExponent` returns a real number for every arbitrary real function. It cannot represent an infinite exponent. The companion checks that this real-valued $\liminf$ even evaluates to 0 when its argument tends to $+\infty$ on the pinned Mathlib version. More concretely, $\text{Real.log } 0 = 0$, so a tail that is eventually zero produces normalized log-ratio 0 and hence exponent 0. This misclassifies bounded-support distributions if the definition is used indiscriminately.

A finite-exponent predicate with eventual strict positivity is safer for this development. For broader tail classes, use an extended-real convention that assigns $-\log 0 = +\infty$. Do not merely change the codomain while retaining $\text{Real.log } 0$.

Scope

$Cx^{(-\alpha)}$ is an eventual tail model; with arbitrary C it is not a valid survival function on all real inputs. A full Pareto law needs a lower endpoint and a piecewise survival function. The theorem as an asymptotic identity remains sound, but its name should not conceal the difference between the limit theorem and a general tail-exponent API.

Transport of a finite tail exponent

True analytic implication; one Lean direction error and a misleading predicate name.

Proof08_SubexponentialTailExponent.lean:23-66

If $S_2/S \rightarrow 2$ and the log-ratio for S converges to a finite α , then $\log(S_2/S)/\log x \rightarrow 0$. Subtracting that vanishing correction proves the same limit for S_2 . The idea is correct. It works more generally for any finite nonzero limiting ratio, with suitable eventual nonzero assumptions.

Mandatory proof correction

```
-- Submitted final line uses esplit in the wrong direction.
exact Tendsto.congr' e.symm
  (Tendsto.congr' esplit.symm hsum)
```

esplit equates the unsplit expression to the sum. hsum proves convergence of the sum, so esplit must be reversed before transferring convergence to the unsplit expression. [R2]

Lean result

FAIL at line 66: the compiler explicitly requires the reverse eventual equality. The repaired ratio-transfer proof compiles. The generalized companion lemma allows a finite nonzero ratio limit k instead of the hard-coded value 2.

Subexponentiality has not been encoded

The definition is only a relation between two arbitrary functions. It does not assert that either is a survival function or that S_2 is the convolution tail of two independent copies. For a concrete diagnostic, choose an exponential survival S and a shifted exponential survival T that equals $2S$ eventually. The predicate holds, while the actual sum of two unit exponentials has tail $(1+x)e^{-x}$, whose ratio to S tends to infinity.

Rename the relation to describe tail-ratio convergence, and define distributional subexponentiality separately. The “one large summand” interpretation is additional probability theory. The hS_2 nonzero hypothesis can be derived from the ratio’s positive limit and eventual positivity of S , so it is redundant rather than missing.

Convexity of the negative-exponent map

Intended theorem true; elaborated statement false, with additional proof errors.

Proof09_ConvexOnRpowNeg.lean:24-43

For $c > 0$, $c^{-\alpha} = \exp(-\alpha \log c)$. This is the exponential of an affine function, hence convex. Equivalently, its second derivative is $(\log c)^2 c^{-\alpha} \geq 0$. The result is strictly convex exactly when $c \neq 1$; for $c = 1$ it is constant.

Statement and proof corrections

Because `Set` is not open, `bare univ` becomes an implicit variable of type `Set ℝ`. The actual declaration therefore claims convexity on an arbitrary set. Take $c = 1$ and the set $\{0, 2\}$: it is not convex, so `ConvexOn` fails even for the constant function. This is a third false statement, introduced by implicit-name capture. A separate companion diagnostic kernel-checks both the captured signature and this counterexample. Write `Set.univ` explicitly and consider `set_option autoImplicit false`.

`haff` proves `ConvexOn` of a real-valued function. It is not an `AffineMap` object. The call `hexp.comp_affineMap haff` therefore passes a proposition where the API requires structured affine-map data. Construct an actual affine map, or apply the convexity inequality for `exp` directly and normalize the affine arguments. Qualify `Set.univ`. [R4]

Lean result

FAIL with four reported errors. The compiler context contains an implicit `univ : Set ℝ`, explaining why `convexOn_exp` and `convex_univ` do not match. The affine argument has the wrong type, and the intermediate ring sequence also leaves a reflexive inequality unsolved. The `Set.univ` direct-Jensen replacement compiles cleanly.

The companion uses a direct Jensen-inequality proof that is checked on the audit baseline. On newer Mathlib versions an existing `convexOn_rpow_left` lemma may make this a short application after rewriting $c^{-\alpha}$ as $(c^{-1})^{\alpha}$. That newer convenience lemma is not assumed in the pinned repair.

What this does and does not establish

This convexity lemma is a useful ingredient for parameter-mixture arguments. It does not prove an expected-shortfall formula, existence of the relevant moments, a specific gamma/lognormal mixing calculation, or a strict Jensen inequality without nondegeneracy conditions. The reference to four Chapter 21 propositions should be described as motivation, not as four completed formalizations.

Fréchet max-scaling identity

The raw identity is true, including its artificial $n = 0$ case; the CDF label is too broad.

Proof10_FrechetMaxstable.lean:26-47

For positive x and ξ , $(n^{(-\xi)x})^{(-1/\xi)} = n x^{(-1/\xi)}$. Exponentiation gives $F(x)^n = F(n^{(-\xi)x})$. This is the correct scaling direction for a maximum. The algebra is useful.

Lean result

PASS unchanged, with no warnings. The raw formula theorem is valid. The companion additionally verifies the piecewise-zero CDF formula identity for every real x and every positive natural n .

The global definition is not a Fréchet CDF

As written, `frechetCDF` $\xi \ 0 = 1$ because the zero-base negative powers are totalized. With $\xi = 1$, its value at $x = -1$ is e , exceeding 1. The file's support caveat recognizes part of this issue, but allowing $n = 0$ sends the right-hand argument to 0 even when the input x is positive.

Importantly, this does **not** refute the written algebraic theorem: its $n = 0$ case is $1 = 1$ under the raw definition. After correcting the CDF to be zero on $x \leq 0$, however, the same $n = 0$ statement becomes $1 = 0$. A genuine CDF theorem must require $n > 0$.

Checked repair in the companion

```
noncomputable def frechetCDF (xi x : Real) : Real :=
  if 0 < x then Real.exp (-(x ^ (-1 / xi))) else 0

-- Correct global statement: xi > 0, n > 0, any real x.
-- (frechetCDF xi x)^n =
--   frechetCDF xi ((n : Real)^(-xi) * x)
```

This proves the corrected formula identity, not the construction of a probability measure or a maximum-of-independent-variables theorem. “Standardized GEV” also needs a stated coordinate convention; the usual location-zero, scale-one GEV uses $1+\xi x$ inside the power.

Gumbel max-shift identity

Correct statement, correct sign, and appropriate positive-sample-size hypothesis.

Proof11_GumbelMaxstable.lean:22-37

The calculation is $\exp(-(x - \log n)) = n \exp(-x)$, followed by $\exp(a)^n = \exp(na)$. Therefore $G(x)^n = G(x - \log n)$. The negative shift is correct: the maximum itself is shifted to the right by $\log n$, so its CDF evaluates the original G at $x - \log n$.

Lean result

PASS unchanged, with no warnings. The companion retains the argument. This is a fully correct, clean analytic lemma as delivered.

Edge cases and credit

The hypothesis $n > 0$ guarantees that $\log n$ and $\exp_log$ are used in the intended positive regime. It also excludes $n = 0$, where the formula with Lean's $\log 0$ would fail. No mathematical correction is needed. The given proof is already short and transparent; further shortening would be cosmetic.

Formalization boundary

The theorem is an exact equality of explicitly defined real functions. To conclude a statement about the maximum of independent Gumbel random variables, add the independence-to-CDF-product result and connect `gumbelCDF` to the distribution. No claim about convergence of maxima from other distributions, or a domain of attraction, follows from this identity alone.

This distinction should be documented without downgrading the identity itself. As an elementary analytic lemma, it is a successful piece of the proposed machinery.

Gaussian specialization of the stable expression

False as written: the location and scale arguments are interchanged.

Proof12_StableCharFnGaussian.lean:26-29 and 33-40

The definition takes arguments in the order $\alpha, \beta, \mu, \sigma, t$. The theorem calls `stableCharFn 2 0  $\sigma$  0 t`. This sets the location to σ and the scale to zero. Its left side is $\exp(i\sigma t)$, the characteristic function of a point mass, not a centered Gaussian.

Decisive counterexample

Set $\sigma = t = 1$. The left side has complex norm 1. The right side is the positive real number $\exp(-1)$, with norm strictly below 1. Thus no correct proof can establish this exact statement. This is a mathematical error, not an uncertain API issue.

Corrected specialization

```
theorem gaussian_specialization {sigma t : Real} :
  stableExpr 2 0 0 sigma t =
    Complex.exp (-(sigma * t)^2), 0) := by
  simp [stableExpr, mul_pow]
```

The essential correction is `2 0 0  $\sigma$` , in that order. The companion's exact theorem proves equality to `Complex.exp (-( $\sigma$ *t)^2), 0)` by simplification. A stronger checked variant permits arbitrary β and μ , yielding imaginary part μt because $\tan \pi = 0$.

Lean result

FAIL: simplification leaves unsupported goals $\sigma = 0 \vee t = 0$ in both coordinate branches. The parameter-corrected specialization compiles, as does a formal proof that the original equality fails at $\sigma = t = 1$.

Parameter meaning

For the repaired expression the variance is $2\sigma^2$. The scale may be any real in this algebraic specialization because it is squared; a distribution API should make its scale convention explicit. Named parameter records would prevent this class of argument-order mistake.

Gaussian scaling under powers

Mathematically correct; zero sample size is also a valid algebraic case.

Proof13_GaussianStable.lean:25-39

The identity is $\exp(-(\sigma t)^2)^n = \exp(-((\sqrt{n})\sigma t)^2)$. Since n is a natural number, $\sqrt{n}$ is defined in the intended nonnegative regime and $(\sqrt{n})^2 = n$. The proof reduces the equality to this square identity and ring normalization.

Lean result

PASS unchanged. The compiler only warns that `push_cast` does nothing. Removing that tactic, or using the shorter companion proof, gives a warning-free result.

Credit and edge cases

No positivity assumption on σ is needed for the literal identity: σ appears squared. $n = 0$ gives 1 on both sides, matching the constant characteristic function of the zero sum if one later supplies the probability interpretation. The sign and square-root normalization are correct.

The notation $-(\sigma * t)^2$ is the negative of the square in the intended Lean parsing. The audit's repair uses extra parentheses for readability; it does not reinterpret it as the square of a negative number or invent a sign error.

Possible simplification

One could use `sqrt` notation throughout rather than converting $n^{(1/2)}$ to `sqrt` only inside the proof. The current two-step reduction is already readable. A generalized complex characteristic-function version would integrate more directly with later probability machinery, but is not necessary to validate this real exponential identity.

The file does not repair Proof12's incorrect parameter call automatically, and it does not itself prove the sum theorem for random variables. Those links must be added explicitly.

Cauchy specialization of the stable expression

False as written for the same parameter-order reason as Proof12.

Proof14_StableCharFnCauchy.lean:25-28 and 32-39

The call `stableCharFn 1 0  $\sigma$  0 t` again gives zero scale and location σ . The left side is $\exp(i\sigma t)$. The target $\exp(-\sigma|t|)$ is a centered Cauchy-form expression with positive scale. For $\sigma = t = 1$ their norms are respectively 1 and $\exp(-1)$, so the equality is false.

Mandatory correction

```
-- Correct argument order:
stableExpr 1 0 0 sigma t
```

```
-- Correct target:
Complex.exp { re := -(sigma * |t|), im := 0 }
```

The companion proves this specialization with $\sigma \geq 0$, a small useful generalization: $\sigma = 0$ gives the degenerate point mass. If the name is reserved for a nondegenerate Cauchy law, retain $\sigma > 0$ in the public probability theorem.

Lean result

FAIL: simplification leaves unsupported goals $\sigma = 0 \vee t = 0$. The corrected specialization with nonnegative scale compiles, and the counterexample at $\sigma = t = 1$ is formally proved.

The $\alpha = 1$ subtlety

The symmetric $\beta = 0$ specialization is recoverable because the skew term vanishes. It does not validate the tangent-only formula for skewed $\alpha = 1$ laws. In Lean, $\tan(\pi/2)$ equals 0 by totalized division, so the original definition suppresses skewness for every β at $\alpha = 1$. The repair of this special case and the repair of the general definition are separate tasks. [R5]

A claim that multiplying zero by an undefined tangent resolves the classical formula is not the right justification. Define the stable expression with its proper piecewise branch, or clearly state that only the symmetric specialization is under consideration.

Cauchy-form exponential scaling

Correct algebraic identity; distribution terminology requires a scale restriction.

Proof15_CauchyStable.lean:23-28

This is an immediate application of $\exp(a)^n = \exp(na)$, with $a = -\sigma|t|$. Associativity of multiplication gives the target $\exp(-(n\sigma)|t|)$. The proof is simple and mathematically complete as a real exponential identity.

Lean result

PASS unchanged, with only a no-op push_cast warning. Both the cleaned original argument and the shorter exponential-law application compile without warnings.

Short checked replacement

```
simp [mul_neg, mul_assoc] using
  (Real.exp_nat_mul (-(sigma * |t|)) n).symm
```

The original small proof is already reasonable. This simplification is optional; it merely avoids a separate congruence step and cast normalization.

Why the unrestricted σ is acceptable algebraically

For $\sigma < 0$ the formula remains a true exponential equality, but $\exp(-\sigma|t|)$ exceeds 1 when $t \neq 0$ and therefore cannot be the characteristic function of a probability measure. Any claim about Cauchy laws should require positive scale, or nonnegative scale if a degenerate law is admitted.

This is a scope correction to the docstring, not a counterexample to the theorem. $n = 0$ is likewise harmless as an algebraic case. To formalize Cauchy stability under independent sums, combine the valid characteristic function with independence and uniqueness of laws.

General scaling of the tangent-only expression

The expression identity is true; its “general stable law” interpretation is false.

Proof16_StableCharFnStable.lean:4-11 and 27-56

For $\alpha > 0$, multiplying σ by $n^{1/\alpha}$ multiplies $|\sigma t|^\alpha$ by n . Multiplying μ by n scales the location phase similarly. The tangent factor is unchanged. This proves the intended algebraic identity even for parameter values that describe no probability distribution.

Lean result

PASS unchanged, with a minor tactic-style warning. The companion checks both this tangent-only expression identity and the corrected piecewise S1 expression identity without warnings. Passing this theorem does not validate its probability-law docstring.

Missing stable-law constraints

A probability-level interface should restrict $0 < \alpha \leq 2$, $-1 \leq \beta \leq 1$, and $\sigma \geq 0$ (or $\sigma > 0$ for nondegeneracy). At $\alpha = 1$ the skew phase needs a logarithmic branch. The book itself distinguishes these cases. [R1] The companion renames the old function `stableExpr` and states its power identity with explicitly analytic scope.

Concrete refutation of the unrestricted interpretation

At $\alpha = 4$, $\beta = \mu = 0$, $\sigma = 1$, the expression is $\phi(t) = \exp(-t^4)$. For points $(-1/2, 0, 1/2)$ and coefficients $(1, -2, 1)$, positive definiteness would require:

$Q = 6 - 8 \cdot \exp(-1/16) + 2 \cdot \exp(-1) \geq 0$.
 Q is approximately -0.77954 .

In fact $\exp(-1/16) > 15/16$ and $\exp(-1) < 1/2$, so $Q < -1/2$. This violates a necessary condition for a characteristic function. The original expression’s scaling identity can be true while the distribution claim is impossible.

For $\alpha = 1$, Mathlib’s $\tan(\pi/2) = 0$ makes the submitted formula independent of β . That loses all skewed one-stable laws. Correcting the domain alone does not repair this missing branch.

Correct stable-law direction

A piecewise S1 expression

The following is the design used in the companion. It is a definition of an expression, not a theorem asserting existence of a probability measure:

```
noncomputable def stableS1Expr
  (alpha beta mu sigma t : Real) : Complex :=
  let r : Real := |sigma * t| ^ alpha
  let phase : Real :=
    if alpha = 1 then -(2 / Real.pi) * Real.log |t|
    else Real.tan (Real.pi * alpha / 2)
  Complex.exp {
    re := -r,
    im := mu*t + r*beta*phase*Real.sign t }
```

With nonnegative scale and $\alpha = 1$, the phase becomes $\mu t - (2\beta\sigma/\pi)t \log|t|$. At $t = 0$, the overall expression is 1; the apparent logarithmic singularity is multiplied by the vanishing radial/sign term. The probability interpretation and its continuity need explicit theorems.

Do not add an incorrect centering correction

For this S1 parameterization, convolution of n identical laws has location $n\mu$ and scale $n^{(1/\alpha)}\sigma$, including the correctly defined $\alpha = 1$ case. A logarithmic location correction appears when comparing to affine scaling of a single skewed one-stable variable, or when changing parameterization. Those are different operations. The correct criticism of Proof16 is its definition and scope, not a blanket demand for an $n \log n$ term.

Remaining formal targets

1. Construct or import a probability law for valid parameters and identify its characteristic function.
2. Prove the product identity for characteristic functions of the relevant independent sum.
3. Use uniqueness of probability measures from equal characteristic functions.
4. State the resulting law-of-sum theorem with the declared parameterization and degeneracy convention.

The companion includes a parameter record with the domain restrictions. This reduces argument-order mistakes and separates mathematical admissibility from algebraic manipulation. It intentionally does not insert an unproved existence assertion as an axiom.

Relationship to the book

Reference: arXiv:2001.10488v4, third revised edition (2025). Page pairs below are printed page / one-based PDF page. The archive is most accurately described as candidate analytic machinery for a larger formalization. [R1]

Files	Closest source anchor	Pages
01-06	§2.2.1, Eq. (2.1)	9 / 23
07	§5.1; compare §2.2.1	97 / 111
08	§2.2.12, Eq. (2.7)	13-14 / 27-28
09	§21.2.2; §§21.5-21.6, Props. 21.1-21.4	381, 385-386 / 395, 399-400
10-11	§9.1, Eq. (9.1), displayed CDFs	172-173 / 186-187
12-16	§7.2.1, Eq. (7.2); §8.3, Eq. (8.4)	140, 159 / 154, 173

What the supplied files actually cover

Proof01-06 establish local ratio-limit facts. Proof07 is an exact-power example. Proof08 transports a finite limit under a ratio premise. Proof09 proves one convexity ingredient. Proof10-11 prove formula identities. Proof12-16 manipulate exponential expressions. None contains random variables, independence, integrals defining a characteristic function, or an identification theorem for probability measures.

These are reasonable building blocks. They should be tracked as such, with separate rows for completed analytic lemmas and for the book-level results that depend on them. A file-per-proof organization does not establish one-to-one coverage of sixteen book proofs.

Concrete provenance improvements

For every intended book result, record the edition, page, equation or proposition, the exact formal statement, its definitions, all additional assumptions, and the gap between the formal conclusion and the source claim. Keep a theorem-to-source index distinct from a file inventory. Correct Proof07’s log-slope anchor, and identify which Chapter 21 conclusion Proof09 is intended to support.

Where the source is inaccurate, document an explicit corrected statement and its justification. Lean should make such a repair visible rather than conceal it in a changed definition.

Source caveats for future formalization

These are isolated checks prompted by the package's references, not additional failures of the submitted Lean files or a full-book audit. The cited formulas were checked against rendered pages. The derivations below are independent audit calculations. [R1]

Slow variation does not force a constant limit

The discussion near §2.2.1 needs care: $L(x) = \log x$ is slowly varying and diverges. A finite positive limiting value is an extra assumption. Proof01 is itself a rigorous counterexample to conflating these properties.

Equation (21.12): the general shifted-lognormal formula

For the displayed mixing density, take $b = 2$, $\alpha_0 = 3$, $\sigma = 1$, $\lambda = 1$. Then $\alpha = 2 + Z$ with positive lognormal Z . Every conditional Pareto mean $\alpha/(\alpha-1)$ lies strictly between 1 and 2, so the mixture mean does too. The printed expression evaluates to $1 + e$, which exceeds 2. This is a nondegenerate counterexample; no limiting or numerical argument is needed.

The correct general expression is $\lambda(1 + E[1/(b-1+Z)])$. For $b = 1$ it simplifies to $\lambda(1 + \exp(\sigma^2)/(\alpha_0-1))$ under the specified mean/log-variance convention. That special case should not be extended to arbitrary $b \geq 1$.

Equation (21.14): baseline, scale, and finiteness

The displayed equality repeats the same primed expectation on both sides and adds a positive correction. Independently, let $m = \alpha_0 - 1$ and $\alpha - 1$ have gamma shape m^2/s^2 and scale s^2/m . For $m^2 > s^2$:

```
E[1/(alpha-1)] = m/(m^2-s^2)
mixture mean = lambda*(1 + m/(m^2-s^2))
baseline mean = lambda*(1 + 1/m)
excess = lambda*s^2 / (m*(m^2-s^2))
```

Thus the baseline needs correction, and the excess needs λ unless unit scale is explicitly assumed.

Example: $\alpha_0 = 3$, $s = 1$, $\lambda = 2$ gives mixture mean $10/3$, baseline 3, excess $1/3$. The scale-free excess would be $1/6$. If $m^2 \leq s^2$, the inverse moment is not finite; $\alpha > 1$ almost surely is not sufficient.

Repair order and acceptance criteria

1. Establish one buildable project

Pin Lean and Mathlib; centralize definitions; remove duplicate global declarations; give each module a namespace or proper shared imports. Compile both the individual modules and an umbrella import. Record the exact commit and archive checksum beside every verification claim.

2. Fix false statements before tactics

Correct the μ/σ order in Proof12 and Proof14 and the unintended implicit set in Proof09. Add named parameter fields or explicit argument comments. Decide whether `stableCharFn` is an analytic helper or a probability-law interface. If the latter, supply the $\alpha = 1$ branch and admissible parameter restrictions.

3. Repair the proof scripts

Apply the checked companion's changes to composition, power continuity, exponent order, congruence direction, convexity, and any baseline-specific tactic failures. Compile the exact export bytes again after extraction. Do not infer exported-file correctness from the parent project's earlier build.

4. Repair semantic interfaces

Use a valid global Fréchet CDF with $n > 0$ in its max theorem. Rename the two-function ratio predicate. Give finite tail exponents an explicit domain, and design an extended-real version before using light-tailed or bounded-support examples. State positivity and regularity conventions for regular variation.

5. Add book-level probability theorems

Connect tails by convolution and exponentials by characteristic functions. Add the independence and uniqueness steps. Treat Chapter 21 moment calculations separately from elementary convexity, and repair source errors explicitly where necessary.

A credible next handoff

Require: a pinned Lake project; no explicit placeholders or new axioms in proof modules; a successful umbrella build; per-theorem axiom output; exact source mappings; and small boundary checks for $n = 0$, $t = 0$, $\alpha = 1$, $\alpha = 2$, $\sigma = 0$, and invalid scale/index values. The goal is targeted verification of the identified risks, not indiscriminate test volume.

The companion repairs preserve the useful work and make progress concrete. They do not warrant declaring the book formalized. The appropriate next milestone is a reproducibly checked analytic foundation with honest definitions and coverage claims.

Compiler evidence and companion contents

Original files were checked independently with the original imports. The table below summarizes the principal emitted diagnostics; complete logs in the companion are authoritative when a file reports several errors.

Proof	Principal build evidence
01	PASS; no warnings.
02	FAIL L33: simp made no progress.
03	FAIL L33: composition/filter type mismatch.
04	PASS; no warnings.
05	FAIL L50: unavailable rpow method; controlled probes also expose wrong continuity premise and exponent order.
06	FAIL L54, L71: ring runs after field_simp has closed the goal.
07	PASS; unused hα warning only.
08	FAIL L66: esplit has the reverse of the required orientation.
09	FAIL: implicit arbitrary set, wrong affine-map argument, and unfinished inequality.
10	PASS; no warnings.
11	PASS; no warnings.
12	FAIL L39-40: unsolved $\sigma = 0 \vee t = 0$; statement false.
13	PASS; no-op push_cast warning only.
14	FAIL L38-39: unsolved $\sigma = 0 \vee t = 0$; statement false.
15	PASS; no-op push_cast warning only.
16	PASS; minor tactic-style warning only.

The three repair modules contain 29 checked theorems, including stronger/simpler variants and diagnostics. Every printed theorem axiom list contains only propext, Classical.choice, and Quot.sound, with no sorryAx. These ordinary foundational dependencies are expected; they are not new axioms introduced to bypass a proof. An additional implicit-set diagnostic verifies Proof09’s elaboration error and its counterexample. All four modules build together successfully.

The repair code has three main namespaced modules for regular variation, tails/extreme values, and stable expressions, plus an implicit-set diagnostic module. Code shown in this PDF is either an exact short replacement or explicitly marked as schematic; use the companion files for copy-and-build execution.

References and audit provenance

[R1] Nassim Nicholas Taleb, Statistical Consequences of Fat Tails, third revised edition, arXiv:2001.10488v4 (17 September 2025). [Open source](#).

[R1 PDF] Versioned source PDF; the relevant pages were locally rendered and checked. [Open source](#).

[R2] Mathlib v4.24.0, filter limits: Tendsto.comp and Tendsto.congr'. [Open source](#).

[R3] Mathlib v4.24.0, real-power continuity and Tendsto.rpow_const. [Open source](#).

[R4] Mathlib v4.24.0, convex functions and ConvexOn.comp_affineMap. [Open source](#).

[R5] Mathlib v4.24.0, real trigonometric functions and tan_pi_div_two. [Open source](#).

[R6] Mathlib v4.24.0, real-power definitions and identities. [Open source](#).

[R7] Mathlib v4.24.0, exponential power identities. [Open source](#).

Source artifacts

Submitted artifact: mathlib-proofs-16.zip. ZIP hash is recorded at the start of this report. All original-file references use the line numbering in the verified archive. Companion repair names intentionally distinguish replacements from original proofs.

Inspected book PDF SHA-256:

```
758e18b7337840104db93296144d67cf551  
4bf2de128fe557dc84bc32a0ad567
```

Audit date: 18 September 2026. The verdicts rest on exact submitted source, explicit mathematical counterexamples, the pinned compiler run, and direct checks of the relevant source formulas. Moving online Mathlib documentation was used for discovery; pinned source and actual compiler results govern the executable claims.